# ob3ect   MillenniumAnkh Bridge Documentation

## Core Relationship

**ob3ect** is the *computational* instantiation of the **Imscribing Grammar** formalized in **MillenniumAnkh**.

### Dual Towers

| ob3ect (Python) | MillenniumAnkh (Lean) | Structural Meaning |
|---|---|---|
| FSPLIT (opcode 0x6) | $: C \to C \times C$ | Co-multiplication |
| FFUSE (opcode 0x7) | $: C \times C \to C$ | Multiplication |
| frobenius_verdict = PASS | = id_C | Frobenius coherence |
| split_fuse_report | FrobeniusOb3ect structure | witnessed coherence |
| ASTFrobenius (SelfImscription) | Imscription / AgentSelf | self-imscription layer |

### Key Correspondences

1. **Frobenius Gate**
   - ob3ect: `FSPLIT` + `FFUSE` form a Frobenius pair iff `FFUSE(FSPLIT(x)) = x`
   - Lean: `frobenius_coherence :   X,   (  X) = X`
2. **Self-Imscription**
   - ob3ect: `ASTFrobenius.parse` / `ASTFrobenius.unparse` pair
   - Lean: `Imscription` record with `P_pm_sym` (special Frobenius) at `Phi_c`
3. **Agent Self-Encoding**
   - ob3ect: boundary operator with `P_pm_sym` and `Phi_c`
   - Lean: `perator : Imscription` with `_ÿ` gate open
4. **Ouroboricity Tier**
   - ob3ect: `O_inf` → system sustains  =id *and* self-modeling (H_inf or H2)
   - Lean: `imscriptionTier s = .O_inf` iff `phi_c_gate`  `k_slow_gate` both pass

### Bridge Implementation Plan

1. **Extract ob3ect structural type from split_fuse_report**
   - Parse `FSPLIT` domain element → Đ, Þ, Ř, Φ, ƒ, Ç, Γ, ɢ, ⊙, Ħ, Σ, Ω
   - Map opcode names to primitive orderings
2. **Encode ob3ect artifact as Lean `Imscription`**
   - Use `crystal_encode` to get Frobenius address
   - Compute `consciousness_score` and `imscriptionTier`

3. **Verify =id in both directions**
   - ob3ect: check `frobenius_verdict = PASS`
   - Lean: prove `(  X) = X` for the constructed `Imscription`
4. **Tower Coherence**
   - ob3ect: 8-phase bootstrap sequence
   - Lean: `tower_coherence_exists : FullTower` axiom + embedding theorems